COMPANY NAME/LOGO

Google Directory Integration Service (GDIS)

Structure Guide

Version: 6.2 (and later) Last

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Table of Contents

Introduction **1**
Purpose of this Guide 1
Assumptions 1
Enabling the API for Google Access/Google Suite (G-Suite) **2**
User and/or Group Synchronization **2**
Users 3
Orgunits/Groups 4
Application Installation Details **5**

Revision History

Item #	Release #	Revision Date	Dist/Pub Date	Description	Tracking Notes
000	6.2 (and later)	05/25	05/25/18	Original	29685

Introduction

GDIS integration was developed to enable synchronization of Google users and groups with PRODUCT NAME users and groups.

Purpose of this Guide

This document is intended for technical administrators or staff who want to understand the structure of the GDIS integration with the PRODUCT NAME Platform. Conceptual and methodological information (as well as important notices and warnings) is provided.

IMPORTANT!

As of version 6.1, the name of the "iSynergy platform was changed to "PRODUCT NAME . You may notice some references and revised files paths have been updated to reflect this change.

Please review the 6.1 Release Notes (in online help or via the PRODUCT NAME Learning Library: https://training.PRODUCT NAME.com/lesson/PRODUCT NAME-platform-6-1-release-notes***) for more information about the name change.***

Assumptions

The following must be true in order to successfully configure and implement GDIS:

- ✓ **Server Prerequisites** are installed.
- ✓ The **PRODUCT NAME platform** is installed and successfully tested as http.
- ✓ An **SSL certificate** was purchased and configured for PRODUCT NAME.
- ✓ **Notepad ++** (run as an administrator) is used when editing config files.

IMPORTANT!

- ❖ ***All access within GDIS is read-only.***
- ❖ ***When referencing a "group in Google, it may be referred to as "orgunit or "distribution group . The current implementation of GDIS uses the term "orgunits to map to PRODUCT NAME Groups. However, It is possible that future client-integrations may require use of Google "distribution groups .***

**Go to https://en.wikipedia.org/wiki/Certificate_authority for more information about certificates and how to obtain them.*

Enabling the API for Google Access/Google Suite (G-Suite)

1) Link the API to the account.
2) Create a service account for the API access.
3) Allow access by the service account to various parts of Google (known as "Scopes).
   - *Scopes allow access to users and orgunits in both Read and Read/Write modes.*
4) Use the following to access the Admin Home for Google Suite:
   - ✓ https://admin.google.com/AdminHome
   - ✓ Login Email: devadmin@PRODUCT NAME-dev. com
   - ✓ Login password: COMPANY NAME1

User and/or Group Synchronization

- Both Google users and orgunits have a non-changing, unique ID created in the following way:
  - PRODUCT NAME makes an initial attempt to link with an established ID. However, if the ID does not yet exist, users are associated via email; groups are associated by group name.
  - Once one or both of the associations (described above) are successful, a link is created using the ID stored within the PRODUCT NAME database.
- The link created with the ID is always used for synchronization.

NOTES

- *Synchronization is based on the ID rather than variable information. By using the ID, the user name and/or email can be changed without breaking the link. This is also true of links to group names.*
- *The IDs in Google are externally meaningless strings, and compared in a case-insensitive manner.*
- *Because GDIS only identifies users and groups it creates by CreatedBy = "PRODUCT NAME Google Dir Sync , GDIS ignores users and groups otherwise created in PRODUCT NAME.*

- When synchronizing begins for Google users and groups, PRODUCT NAME users and groups are compiled into lists in the GDIS service. Those lists are then used locally for user and group information (vs fetching individual users and groups): thus making the process more efficient.
- The Google API organizes groups of users: reading a maximum of 500 at a time; PRODUCT NAME reads Google users into hard coded groups of 200. However, orgunits are not grouped (see diagram on page 6).
- PRODUCT NAME uses Attribute tables to create links between Google users and PRODUCT NAME users, as well as for Google orgunits and PRODUCT NAME groups. Each Attribute table has an ID to allow selection of only pertinent attributes.

IMPORTANT!

Throughout the remaining text in this document, "Attribute should be assumed to be scoped to GDIS.

- PRODUCT NAME creates a list from all of current Google Users; it then looks at the UserAttribute table for a value matching the ID and the correct UserAttributeDefintionID. A similar list is created from the list of current Google Groups (i.e., orgunits).
  - If a match is found in the UserAttribute table, it is synched with the corresponding PRODUCT NAME user.

User Details

New Users

- In the case of a new user, since there is likely to be no match (as described on page 2), PRODUCT NAME searches for the new user by email.
- If an exact, case-insensitive match for the email is found, a UserAttribute record is created and linked to the corresponding PRODUCT NAME user.
- If no match is found, a new record is created in the UserAttribute table: creating a new PRODUCT NAME user.

Updated Users

- If a Google user name or email has been changed, it is updated in (and synched with), PRODUCT NAME via the ID (described on page 2).
- Any other changes to a user's Google account have no bearing on its link to PRODUCT NAME.

IMPORTANT!

Only PRODUCT NAME users created by "PRODUCT NAME Google Dir Sync can be updated.

Deleted Users

- In order to maintain referential integrity (as well as audit history), PRODUCT NAME does not delete users from its database; however, Google Directory Services does allow users to be deleted. Therefore, when users are deleted from Google Directory Services, PRODUCT NAME "marks them for deletion" and sets the PasswordExpInterval for user records in PRODUCT NAME to a value of "-99 : indicating that the user is no longer active in PRODUCT NAME, and is unable to log into the system.
- Google Users can be suspended (and un-suspended) or deleted. Once a Google user is deleted" he/she no longer appears in the list in Google.
- Users suspended, unsuspended, and/or deleted from Google are treated equally in PRODUCT NAME (i.e., such users are designated as deleted*.
- If a user is marked as deleted* or suspended, PRODUCT NAME uses the link to the ID (in the UserAttribute table) to locate the user record. If /when a match is found, the user is marked as deleted*.
- If the user is not found, then PRODUCT NAME looks for the user by email. If a match is made using the email, PRODUCT NAME checks the CreatedBy = PRODUCT NAME Google Dir Sync. Once this is verified, the user is marked as deleted*; otherwise the record is ignored.
- Users who have been permanently (hard) deleted no longer show up in the list of users from Google. At the end of user routine syncronization, PRODUCT NAME matches the entire list of Google users against the entire list of PRODUCT NAME users.
- Any user in the PRODUCT NAME list, but not in the Google list...
  - is a candidate for deletion*.
  - is designated by PRODUCT NAME as deleted* when the user shows as CreatedBy = "PRODUCT NAME Google Dir Sync.

**PRODUCT NAME designates users marked for deletion as "-99 .*

Orgunits/Groups

New Groups

NOTE

Only Google orgunits with users are added (i.e., "empty groups are not added).

- As previously noted, PRODUCT NAME creates a list of all orgunits from Google; orgunits become groups in PRODUCT NAME.
- A Google user can be a member of only one Google orgunit; however, orgunits can be nested (see diagram on page 6).
- Hierarchical groups are supported in Google, but not in PRODUCT NAME; therefore, PRODUCT NAME "flattens the tree structure to convert the hierarchy in to equal groups (see diagram on page 6).

EXAMPLE

A Google orgunit of "Accounting with three sub-Organizations "payroll , "AR and "AP is converted to four groups in PRODUCT NAME: "Accounting", "Accounting - payroll , "Accounting - AR and "Accounting - AP .

- Orgunits and sub-orgunits are separated by a space hyphen space or "X – Y".
- Once PRODUCT NAME recognizes a list of all orgunits from Google, it looks for a UserGroupAttribute record with an AttributeValue that matches the orgunitID, along with the UserGroupAttributeDefintionID.
  - If a an ID match is not found, PRODUCT NAME looks for an exact, case-insensitive match in the group name.
  - If an exact, case-insensitive match by group name is found, a UserGroupsAttribute record designated as CreatedBy to "PRODUCT NAME Google Dir Sync is created.
  - If a matching group name is ***not*** found, a new PRODUCT NAME group and UserGroupAttribute record are created.

New and Deleted Group Users

New Users

- New users in a Google orgunit are added to PRODUCT NAME using the attribute table to get the PRODUCT NAME UserID; PRODUCT NAME users not already in the group are added.
- It is a safe assumption that if PRODUCT NAME users are listed in the attribute table, they were created by GDIS; therefore, ***NO*** verification check is required as it is with individual users.

Deleted Users

- When users are removed from a Google orgunit, they are also removed from the associated PRODUCT NAME group.
- PRODUCT NAME exclusively checks the attribute table for linking, but does not make a CreatedBy verification.

Deleted Groups

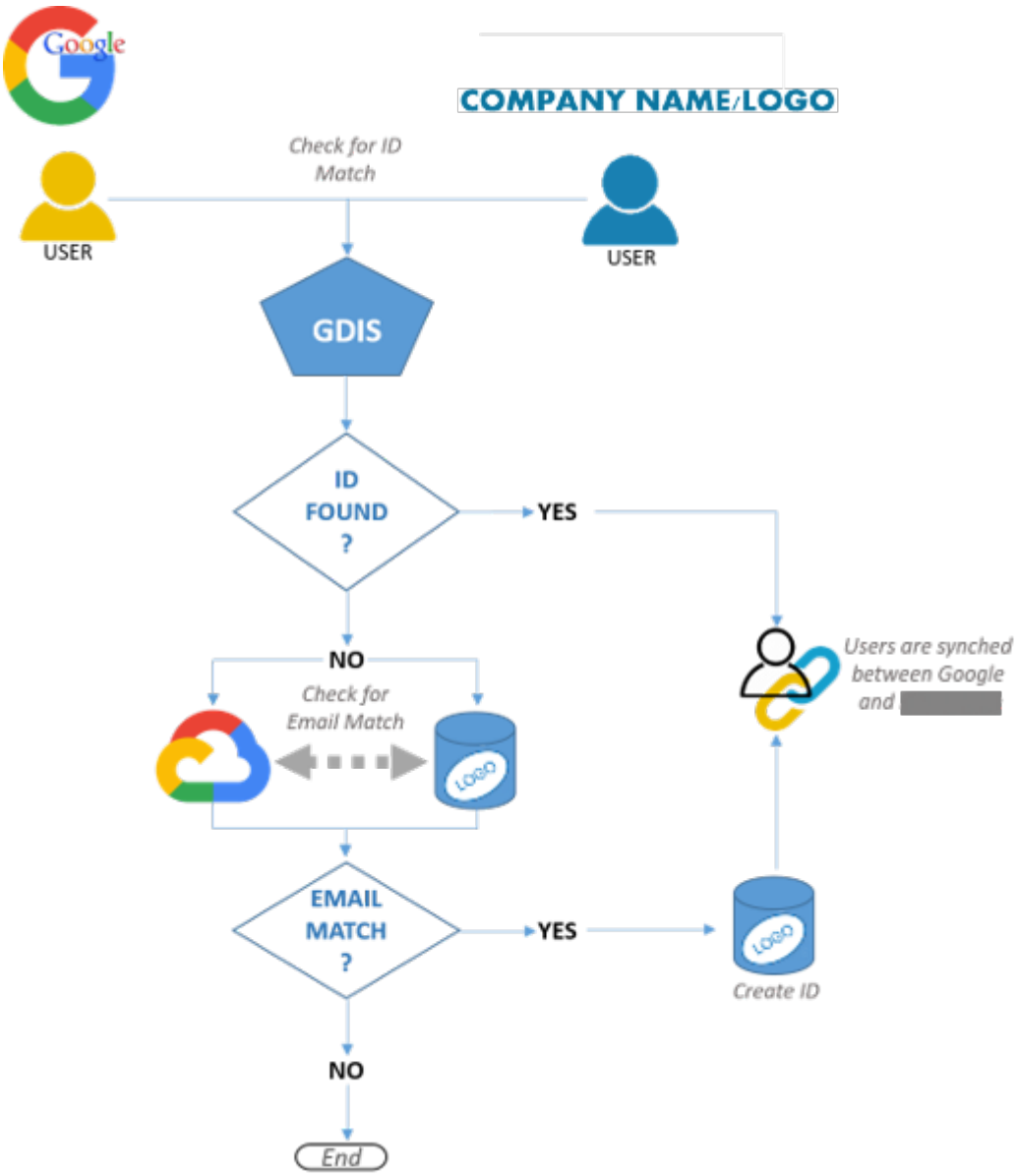
- Any PRODUCT NAME group not in the list of Google orgunits is a candidate to be deleted.
- The list of candidates is checked, and only those designated as CreatedBy = "PRODUCT NAME Google Dir Sync are deleted.

Application Installation Details

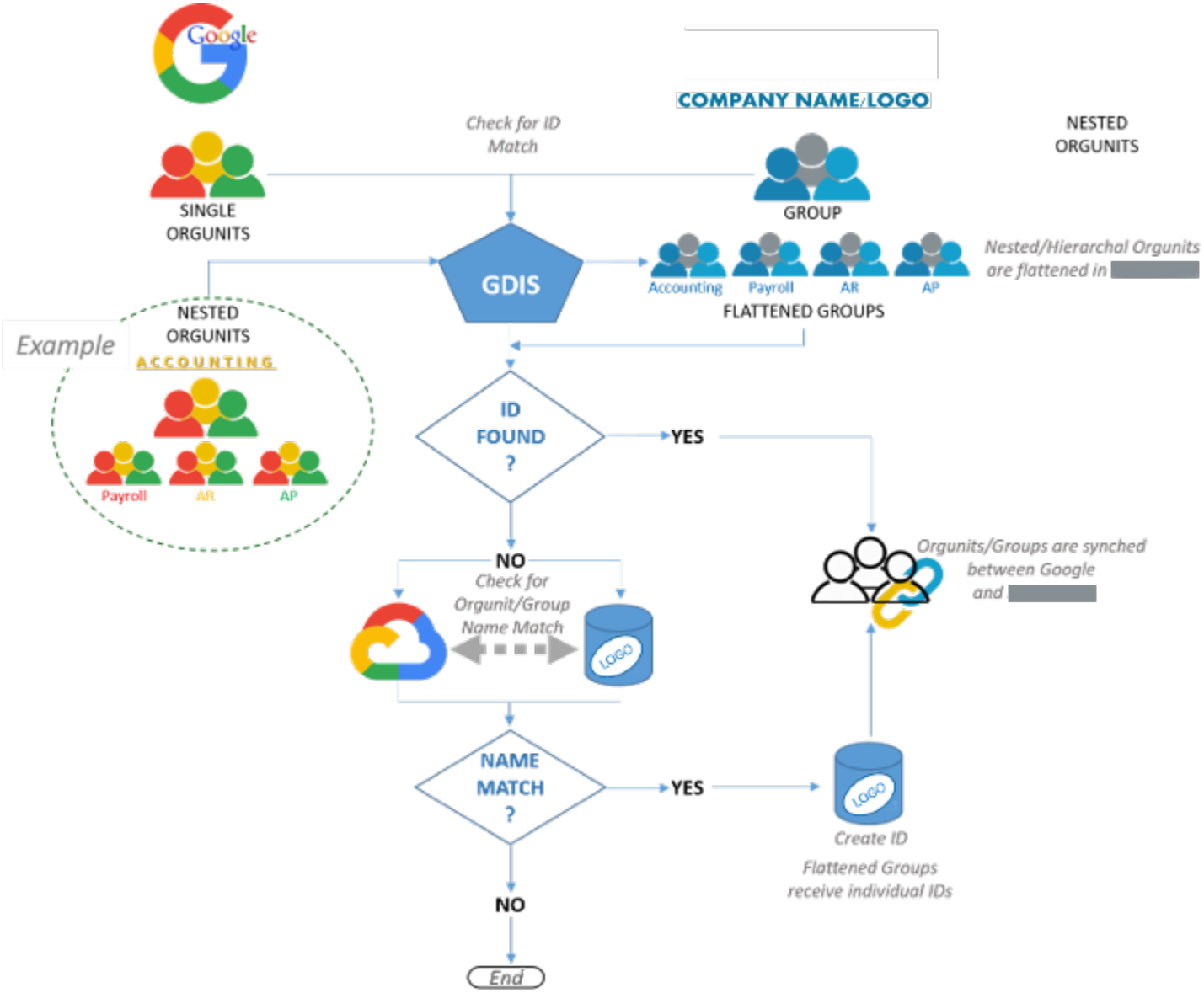
- The application is installed as its own service; it is a 64 bit application and installed to \PRODUCT NAME\programs\gdisservice.
- Frequently, a client may want to run this service on their own servers rather than the PRODUCT NAME servers; since the service communicates with PRODUCT NAME via web services, it can be deployed anywhere.
- Because of the size of PRODUCT NAME Services, GDIS Integration has its own service rather than being included in PRODUCT NAME Services.
- When clients want to install this service on their own server(s), PRODUCT NAME provides them with a 100K install, rather than a 200M.
- The install requires an input of a user ID for running the service: "LocalService", with no password.
- Once the User ID has been entered, the remainder of the settings must be set manually in the PRODUCT NAME.GDISService.exe.config, and are documented in \PRODUCT NAME\programs\gdisservice\ PRODUCT NAME.GDISService.exe.config.

Diagrams of Synchronization Processes

User Synchronization



Diagrams of Synchronization Processes (continued on next page)

Diagrams of Synchronization Processes (continued)

Orgunit/Group Synchronization